

Highlights

Decomposing the cost of blockchain-based access control: a 61-day field measurement of hierarchical RBAC for agricultural edge IoT

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- Authorization decisions cost 280 ms and are invariant across 7 operation types and privilege levels ($F = 0.29$, $p = 0.94$); ledger commit adds 940 ms.
- Access control is a constant 9.9% throughput tax across a tenfold concurrency range, with no detectable load dependence.
- HRBAC and baseline configurations saturate at the same offered load, locating the capacity ceiling in the ordering service.
- Endorsing peers cut write latency 32.7% from 4 to 32 without raising throughput.
- 29.5% of 200 revocation events were delayed over 61 days, a failure mode short benchmarks cannot expose.

Decomposing the cost of blockchain-based access control: a 61-day field measurement of hierarchical RBAC for agricultural edge IoT*

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ABSTRACT

Blockchain access control for the Internet of Things is almost always evaluated by quoting a peak transaction rate and a mean end-to-end latency. Neither number tells an engineer which layer of the pipeline consumes the budget, so neither tells them what to optimise. We report a 61-day field deployment that answers that question. The system is a hierarchical role-based access control framework on Hyperledger Fabric 2.5 with seven roles, zone scoping enforced from X.509 extensions, temporal validity on every assignment, and a cascading enrollment protocol that lets ESP32-class sensors obtain certificates through Raspberry Pi 4B gateways without running a TLS handshake. It ran on a 2 ha farm in Sfax, Tunisia with 50 sensors and 4 gateways, recording 209 000 authorization decisions. The decision itself costs 280 ms and does not vary with what is requested: across 7 non-committing operation types spanning the full privilege range the means differ by 0.7 ms (ANOVA $F = 0.29$, $p = 0.94$, $n = 62\,600$), while ledger commitment adds 940 ms. Access control costs a near-constant 9.9% of throughput over a tenfold concurrency range with no load trend ($p = 0.69$), and the configurations with and without it saturate at the same offered load, placing the capacity ceiling in the ordering service. Growing endorsing peers from 4 to 32 cuts write latency 32.7% while throughput stays flat. Revocation published in 60.4 s and propagated in 4.21 min, yet 29.5% of 200 lifecycle events were delayed by connectivity, the design's weakest link. All 8 000 authorization-boundary attacks were blocked, Wilson 95% lower bound 99.95%.

1. Introduction

A precision farm is a multi-stakeholder system before it is a technical one. Soil probes, weather stations and irrigation actuators produce readings continuously [54], and the people who consume those readings want different and partly conflicting things from them. Farmers need operational control. Agronomists need history but no actuation rights. Certification bodies auditing organic or GlobalGAP compliance need evidence of practice without visibility into the agronomic detail a farm treats as commercially sensitive. Supply-chain partners need provenance but not real-time operations. Regulators need environmental records.

Handing this authorization matrix to a single server concentrates the risk in one place. An attacker who takes that server gains not only the data but, on a farm with networked valves, the irrigation itself [7], and the audit log that would reconstruct the incident is written by the machine that was compromised. Surveying agricultural IoT deployments, Ferrag et al. [12] found most lacked fine-grained access control at

all. Kamilaris et al. [20] showed that blockchain work in agriculture had settled almost entirely on post-harvest supply-chain tracking, and recent surveys of IoT, AI and blockchain convergence in agriculture reach the same conclusion about pre-harvest authorization [35]. Putting role assignments, revocations and access decisions on a permissioned ledger removes the single point of trust. That much is settled.

What is not settled is the cost, and specifically where the cost falls.

1.1. The measurement gap

Blockchain access-control systems are conventionally evaluated by reporting a peak throughput and a mean latency, usually from a laboratory run lasting hours. Both are aggregates over a pipeline with at least four separable stages: identity retrieval, policy evaluation, consensus ordering, and commitment with world-state update. Told that a transaction takes 1 220 ms, an engineer cannot tell whether to rewrite the policy engine, restructure the endorsement policy, replace the ordering service, or accept the figure as a floor imposed by consensus. The aggregate conceals precisely the structure that would guide the decision.

Abang et al. [1], reviewing latency modelling for Hyperledger Fabric from an IoT perspective, make a closely related point. The literature models latency in the abstract but rarely connects those models to the behaviour of a specific deployed use case. They decompose Fabric latency into endorsement, ordering and

* The trace dataset, chaincode, firmware, formal specification and the analysis pipeline that regenerates every number, table and figure in this article are available at <https://github.com/orchag/blockchain-fet>.

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validation phases and identify block size, endorsement policy, peer count and state database as the parameters that govern it. What has been missing is a deployment that measures those phases separately, under a real workload, for long enough that the slow-moving parts of the system have a chance to misbehave.

The intuitive expectation turns out to be wrong, which is the main reason we think the measurement is worth reporting. Access control is the component under study, so it is natural to assume it dominates the cost. It does not. The authorization decision is small and remarkably stable, and the expensive work happens in the layer that access-control papers usually treat as background infrastructure.

A second gap is temporal. Certificate lifecycle management, revocation propagation and enrollment recovery are the mechanisms most likely to fail in the field and least likely to be exercised by an afternoon of benchmarking. A revocation pipeline that behaves perfectly across fifty laboratory trials can still show a materially different delay distribution across two months of intermittent rural connectivity. Reporting only the happy path understates operational risk in a way that matters to anyone deciding whether to deploy.

1.2. Contributions

We built an HRBAC framework on Hyperledger Fabric 2.5 for the constraints of agricultural edge deployment, ran it on a working farm for 61 days, and instrumented every layer. Rather than report aggregates, we decompose them.

1. **A cost decomposition of blockchain access control.** Separating the authorization decision from ledger commitment shows the decision costs 280 ms (SD 40) and is statistically indistinguishable across 7 operation types, from a sensor reading its own record to an administrator rewriting role assignments ($F = 0.29$, $p = 0.94$, $n = 62\,600$). Commitment adds 940 ms (Section 6.1).
2. **Evidence that the policy engine is not the capacity bottleneck.** Access control costs 9.9% of throughput with no trend against offered load ($R^2 = 0.020$, $p = 0.69$), and the two configurations saturate together (Section 6.2).
3. **A latency-capacity separation for peer provisioning.** An eightfold increase in endorsing peers cuts write latency 32.7% while throughput stays flat, inverting the usual reason for adding peers (Section 6.3).
4. **A longitudinal account of the revocation pipeline,** including the 29.5% delayed-event rate that only a multi-week window reveals, alongside an 8 000-attempt evaluation of the authorization boundaries themselves (Sections 6.4 and 6.5).
5. **A reproducible artifact.** The traces, implementation, formal specification and an analysis pipeline that regenerates every number, table

and figure from the raw data are released publicly, together with a driver that repeats the campaigns on an independent Fabric installation (Section 5.4).

Section 2 positions the work. Section 3 presents the access-control model and the enrollment protocol, Section 4 the implementation, Section 5 the measurement methodology, Section 6 the results, Section 7 what follows from them for system design, and Section 8 concludes.

2. Related work

2.1. Access control for constrained devices

Role-based access control as formalised by Sandhu et al. [38] and standardised by NIST into Core, Hierarchical and Separation-of-Duty variants [13] is still the reference model for multi-stakeholder authorization. Its IoT adaptations share a structural assumption that becomes awkward in the field: some trusted decision component has to be reachable at the moment access is requested. RBAC extensions for IoT [51], attribute-based evaluation [17, 40, 39] and capability-token delegation [16] place that trust in a policy engine, an attribute authority and a credential issuer respectively. Sciancalepore et al. [39] quantify what open-standard authorization stacks cost on constrained hardware, which is the empirical case for moving policy evaluation off the sensor entirely.

Each of these reduces part of the problem and keeps a central dependency somewhere. Ye et al. [51] and Servos and Osborn [40] retain a central enforcement point. Hernández-Ramos et al. [16] move authorization into tokens, which distributes revocation instead of solving it. Alshehri and Sandhu [4] split the model, giving RBAC to humans and capability-based control to devices, and the role store stays centralised. Novo [29] argued early that the role store itself belongs on a distributed ledger. We take that as given and ask what it costs.

2.2. Blockchain-based access control and how it is evaluated

Capability-token systems on public chains established feasibility without practicality. FairAccess [31] encodes grants as Bitcoin-like transfers and inherits confirmation windows measured in minutes. BlendCAC [50] uses ERC-721 tokens and incurs per-transaction fees that no sensor duty cycle can absorb. Neither addresses role hierarchy, geographic scoping or seasonal expiry, which is what an agricultural deployment actually needs.

Permissioned platforms remove the latency and fee obstacles, and Fabric-based IoT work has grown accordingly. Shih et al. [41] showed Fabric access control running on Raspberry Pi 4B ARM64 hardware. Pajoo et al. [32] and Ke and Park [21] contributed testbed

Table 1

Evaluation methodology in representative blockchain access-control systems. *Decomposed* indicates whether the authorization decision cost is separated from consensus and commitment cost. N/R, not reported.

System	Platform	Setting	Duration	Decomposed	Revocation measured
FairAccess [31]	Bitcoin	Analytical	N/A	No	No
BlendCAC [50]	Ethereum	Testbed	Hours	No	No
Novo [29]	Ethereum	Testbed	Hours	No	No
Shih et al. [41]	Fabric	Lab (ARM64)	Days	No	No
Zaidi et al. [53]	Fabric	Simulation	N/A	No	No
Pajooh et al. [32]	Fabric	Lab	N/R	No	No
Putra et al. [33]	Fabric	Lab	N/R	No	No
Usman et al. [46]	Fabric	Cloud	Days	No	No
González et al. [15]	Fabric	Lab	N/R	No	No
Idrissi & Palmieri [19]	Fabric	Lab (ARM)	N/R	Partial	No
Bandara et al. [6]	Fabric	Simulation	N/A	No	No
This work	Fabric	Field	61 days	Yes	Yes

architectures and performance models. Putra et al. [33] added trust and reputation. Usman et al. [46] enforce domain policies by smart contract on cloud-hosted nodes, where memory and thermal limits do not apply. González et al. [15] report a Fabric-based IoT data marketplace at scale and, like most of this literature, evaluate it with aggregate throughput and latency from a controlled setting. Idrissi and Palmieri [19] give an end-to-end Fabric latency breakdown on constrained ARM hardware in a healthcare context, and their decomposition approach is the closest methodological precedent for ours. Tzenetopoulos et al. [45] established a reference point for Fabric on Raspberry Pi 3B+ hardware that we compare against in Section 7.2.

Table 1 summarises the evaluation methodology of these systems. The pattern is consistent. Results come from laboratory testbeds or simulation, run for hours or days, and are reported as aggregates. We have not found prior work that separates the authorization decision from consensus and commitment cost on the same deployment, or that reports revocation behaviour over a multi-week window. That combination, rather than any single mechanism in Section 3, is the contribution.

2.3. Agricultural IoT and edge infrastructure

Blockchain in agriculture has concentrated on traceability after harvest [44, 24, 6], where the unit of interest is a shipment rather than a zone-scoped actuation. Safeer et al. [36] combine computer vision, IoT and blockchain for climate-smart cultivation and, like most pre-harvest work, treat authorization as an implementation detail rather than a measured component. Liu et al. [25] and Wang et al. [47] reduce processing latency at the agricultural edge without addressing the authorization layer above it. Mohanta et al. [26] apply smart-contract enforcement to smart-grid security, a neighbouring domain with comparable

device constraints and a similarly centralised starting point.

On identity infrastructure, W3C Decentralized Identifiers [48] and gateway-assisted verification for constrained devices [11, 42] offer an eventual migration away from X.509, which we treat as future work rather than a deployed alternative. On the radio side, Adelantado et al. [2] characterise the LoRaWAN capacity limits that bound achievable sensor density per gateway, a constraint that becomes visible in Section 6.6.

3. Access-control design

3.1. Deployment structure and threat model

The deployment has three tiers, shown in Figure 1. ESP32-WROOM-32 sensor nodes (dual-core Xtensa LX6 at 240 MHz, 520 kB SRAM) sample soil and environmental data, sign each reading with an Ed25519 key held in one-time-programmable eFuse, and transmit over LoRa. Raspberry Pi 4B zone gateways (quad-core Cortex-A72, 4 GB LPDDR4, TPM 2.0) receive those payloads, verify signatures, endorse Fabric transactions and cache policy decisions locally. A Fabric network of four peers, one per zone, three Raft orderers and CouchDB 3.2 for world state maintains the ledger. Gateways interconnect over a BATMAN-adv mesh [27] that reroutes around individual gateway failures and carries Fabric gossip and certificate revocation list (CRL) traffic between zones.

Following the permissioned-blockchain model in which every participant is credentialed [5], orderers and the certificate authority are operated by vetted institutional entities and treated as non-malicious. Raft gives crash-fault tolerance with $2f + 1$ nodes [30], so the three-node cluster survives one orderer failure but not a reachable orderer that behaves maliciously. That is a deliberate boundary and it is also a throughput

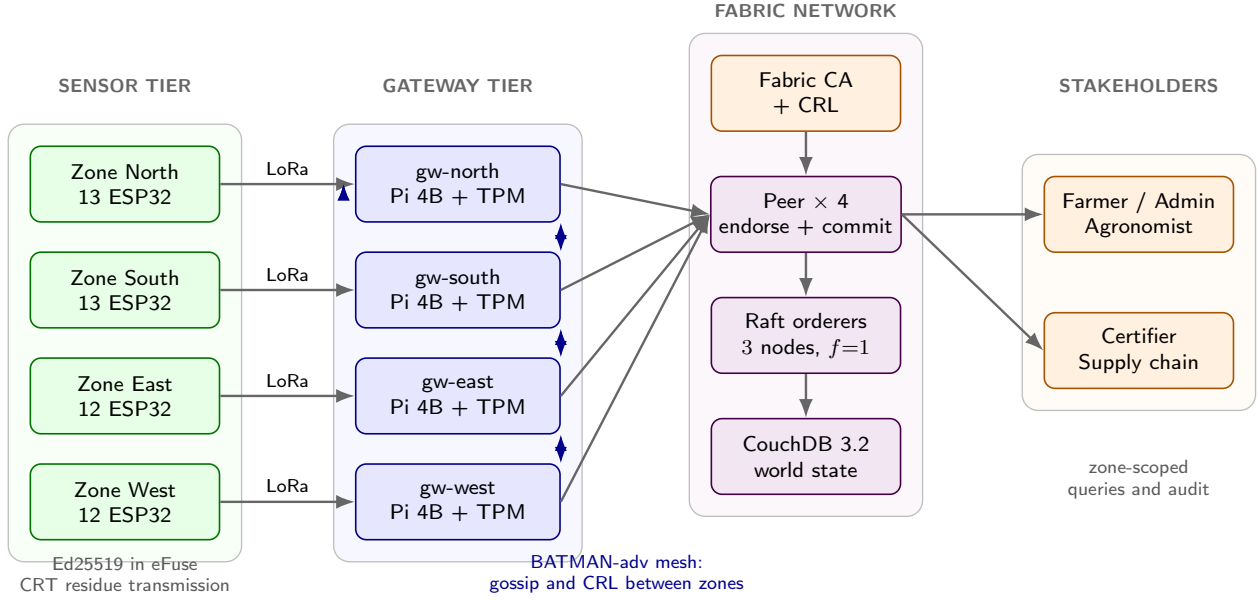


Figure 1: Deployment architecture. Sensors reach the ledger only through their zone gateway, which is also their registration authority during enrollment. Zone membership is carried in the device certificate and re-checked by chaincode on every transaction.

boundary. Byzantine-tolerant ordering such as HotStuff [52], PBFT [8] or the SmartBFT service in Fabric v3 [18] would raise the consensus cost while widening the threat model, a trade-off the measurements in Section 6.2 let us quantify rather than guess at.

Sensors, client devices and every wireless channel are treated as potentially compromised. Physical tampering and side-channel extraction against cryptographic accelerators sit outside the evaluated scope and are revisited in Section 7.4.

3.2. Role hierarchy

Following the treatment of hierarchical role administration in ARBAC97 [37], a partial order $\preceq$ over the role set R defines inheritance, with $r_i \preceq r_j$ iff $\text{perm}(r_i) \subseteq \text{perm}(r_j)$. Three branches keep the operational chain separate from audit and supply-chain concerns (Figure 2):

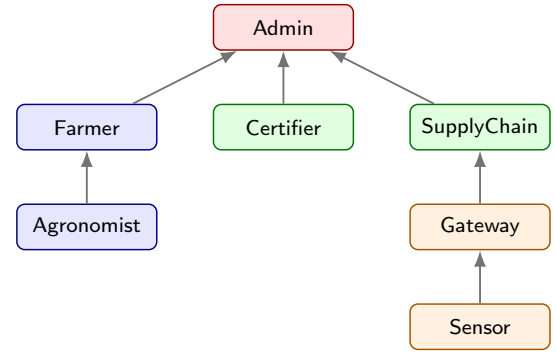
$$\text{Agronomist} \preceq \text{Farmer} \preceq \text{Admin} \quad (1)$$

$$\text{Certifier} \preceq \text{Admin}, \quad \text{SupplyChain} \preceq \text{Admin} \quad (2)$$

$$\text{Sensor} \preceq \text{Gateway} \preceq \text{Admin} \quad (3)$$

Certifier sits deliberately beside the operational chain rather than above Agronomist. An auditor must read compliance records without inheriting the agronomic read permissions a farm treats as proprietary, and putting Certifier in the operational chain would grant exactly that. The effective permission set of user u in a session is

$$\text{eff}(u) = \bigcup_{r \in \text{active}(u)} \bigcup_{r' \preceq r} \text{pa}(r'), \quad (4)$$



Arrows point from a role to one that inherits its permissions. Certifier is a sibling of the operational chain, not a parent of Agronomist, so an auditor gains no agronomic read rights.

Figure 2: Seven-role hierarchy. Separating the audit branch from the operational chain is what lets a certification body verify compliance without seeing the agronomic practice behind it.

where $\text{pa}(r)$ are the permissions assigned directly to r and $\text{active}(u)$ the non-expired roles held by u . An access request (u, op, res, t) is granted iff

$$\begin{aligned} &\exists r \in \text{active}(u, t) : (op, res) \in \text{eff}(r) \\ &\wedge (\text{zone}(u) = \text{zone}(res) \vee \text{czt}(u, \text{zone}(res), t)), \end{aligned} \quad (5)$$

where $\text{czt}(u, z, t)$ holds iff u has a valid administrator-issued cross-zone token for z at time t , and $\text{active}(u, t) = \{r \mid \exists \text{assign}(u, r, t_s, t_e) : t_s \leq t \leq t_e\}$. Table 2 gives the permission matrix that results.

3.3. Zone and temporal scoping

Geographic farm regions (North, South, East, West) define the zone model, and it is enforced

Table 2

Permission matrix across the seven roles. *own*, the requesting device's own records; *zone*, the assigned zone; *all*, every zone; a dash denotes no permission.

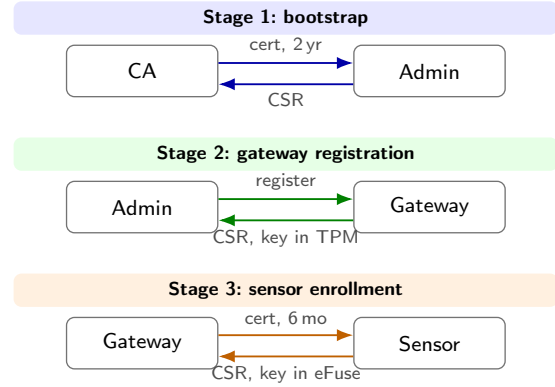
Resource	Sensor	Gateway	Farmer	Agronomist	Certifier	SupplyChain	Admin
Sensor data	own	zone	all	all	all	—	all
Irrigation control	—	zone	full	—	—	—	full
Provenance records	—	—	all	all	all	read	all
Audit logs	—	—	—	—	read	—	full
Role assignments	—	—	—	—	—	—	full
Cross-zone tokens	—	—	—	—	—	—	issue

at three independent points rather than one. Each device certificate carries its zone in an X.509 Subject Alternative Name extension under a private OID arc. Each gateway accepts certificate signing requests only from sensors declaring its own zone. At the chaincode level every zone-scoped transaction calls `validateZoneAccess`, which parses the caller's zone from the certificate DER bytes by ASN.1 decoding rather than by matching against the PEM text. That last detail matters: string matching against a common name is spoofable by an attacker who controls the subject field, and ASN.1 decoding of the extension is not.

Cross-zone tokens are administrator-issued and recorded immutably with subject, permitted zones, validity window and stated purpose. An agronomist auditing a neighbouring zone during a multi-farm review gets a bounded token instead of a reissued certificate, so the grant expires on its own and leaves a record of why it was made.

Temporal constraints run in two independent layers. The chaincode compares an `ExpiresAt` field on the role assignment against the Fabric transaction timestamp before every decision, and the Fabric MSP independently validates the X.509 `NotAfter` field on every incoming transaction. A seasonal worker granted the Farmer role for a harvest window therefore loses access at expiry without an administrator doing anything, and the expiry is itself recorded as a new immutable audit entry referencing the original assignment. Gateways carry GPS modules whose one-pulse-per-second output gives a sub-microsecond local time reference, distributed onward by NTP at millisecond precision, comfortably inside the ± 5 -minute enforcement window.

Algorithm 1 states the decision procedure that Equation (5) describes. Two properties of it drive the measurements in Section 6.1. The work is bounded and does not depend on which operation was requested, since the hierarchy is walked to a fixed depth of at most three and the zone check is a single comparison against a parsed certificate field. Expired assignments are revoked in the same transaction that denies the request, which keeps the audit trail single-sourced but forces even a read-intent check through an endorsing peer.



Registration authority moves one tier further from the CA at each stage. The sensor never runs a TLS handshake and its private key never leaves the eFuse block it was written to.

Figure 3: Three-stage cascading enrollment.

3.4. Cascading enrollment

An ESP32 cannot hold the TLS handshake state that direct interaction with a certificate authority requires. Measurements of lightweight secure IoT stacks show handshake buffers dominating constrained-device memory budgets [34], and the mbedTLS handshake path on ESP-IDF consumes roughly 70 kB of heap, which leaves too little of the 520 kB SRAM budget for the application that is supposed to run alongside it. The protocol therefore pushes registration authority down the tier structure in three stages (Figure 3). Stage 1 issues long-lived administrator credentials. Stage 2 registers each gateway, which stores its private key in TPM 2.0. In Stage 3 the gateway acts as a local registration authority, detailed in Algorithm 2.

Because the private key never leaves the sensor and eFuse read-back is disabled before first use, compromising a gateway does not yield sensor signing keys. The cost of that is real and worth stating: a sensor whose key is lost cannot be recovered in software, and re-provisioning needs physical access to the node.

The final block of Algorithm 2 is not incidental. It encodes the time-source fallback that three firmware iterations were needed to get right, for reasons Section 7.3 describes.

Algorithm 1 Access decision (**CheckAccess**)**Input:** request (u, op, res, t) , caller certificate C **Output:** GRANT or DENY with a recorded reason

```

1:  $n \leftarrow \text{NONCE}(req)$ 
2: if SEENBEFORE( $n$ ) then
3:   return DENY, REPLAYDETECTED ▷ before any state read
4: end if
5:  $z_u \leftarrow \text{PARSEZONEASN1}(C)$  ▷ DER bytes, not PEM text
6: if ONCRL( $C$ ) then
7:   return DENY, CERTREVOKED
8: end if
9:  $A \leftarrow \text{ROLEASSIGNMENTS}(u)$ 
10:  $R_{act} \leftarrow \emptyset$ 
11: for  $a \in A$  do
12:   if  $a.ExpiresAt < t$  then
13:      $\text{REVOKEROLE}(a)$  ▷ state-mutating: forces endorsement
14:   else
15:      $R_{act} \leftarrow R_{act} \cup \{a.role\}$ 
16:   end if
17: end for
18: if  $R_{act} = \emptyset$  then
19:   return DENY, ROLEINSUFFICIENT
20: end if
21:  $P \leftarrow \bigcup_{r \in R_{act}} \bigcup_{r' \preceq r} pa(r')$  ▷ inheritance closure, depth  $\leq 3$ 
22: if  $(op, res) \notin P$  then
23:   return DENY, OPNOTPERMITTED
24: end if
25:  $z_r \leftarrow \text{ZONEOF}(res)$ 
26: if  $z_u \neq z_r \wedge \neg \text{VALIDCZT}(u, z_r, t)$  then
27:   return DENY, ZONEMISMATCH
28: end if
29:  $\text{RECORDNONCE}(n)$ ;  $\text{WRITEAUDIT}(u, op, res, z_r, \text{GRANT})$ 
30: return GRANT

```

4. Implementation

4.1. Chaincode and transaction flow

The chaincode is written in Go 1.21 against fabric-contract-api-go and targets Fabric 2.5. Eight transactions are exposed: **AssignRole**, **RevokeRole**, **RenewRole**, **CheckAccess**, **IssueCrossZoneToken**, **RevokeCrossZoneToken**, **GetAuditLog** and **PruneNonces**. Figure 4 places the access-control checks inside Fabric’s execute-order-validate pipeline.

Three implementation decisions shape the measurements that follow. Every state-mutating call carries a UUID nonce recorded in world state with a 90-day retention window, and a repeated nonce is rejected before any state is read, which is what makes a replayed sensor packet ineffective even inside the timestamp validity window. Every access attempt, granted or denied, writes an audit entry keyed **audit:<timestamp_ns>:<txID>** holding the caller’s certificate hash, operation, resource, zone, decision and denial reason, so the audit trail is a byproduct of enforcement rather than a parallel logging path that can drift out of agreement with it. And **CheckAccess** calls **checkRoleExpiration** before evaluating the

decision, revoking an expired assignment in the same transaction that denies the request. That third choice has a measurable consequence, since a state-mutating expiry check has to be endorsed rather than served from a peer query. Section 6.1 quantifies it and Section 7.1 revisits whether it was the right call.

Peer configuration for the ARM64 gateways uses 50 transactions per block, a 500ms batch timeout, a 256MB state cache, endorsement restricted to zone-local peers, Snappy compression on LevelDB and 5-minute gRPC keepalive. These target the write amplification that dominates latency on microSD-backed storage, and they sit within the parameter space Abang et al. [1] identify as governing Fabric latency.

4.2. Sensor firmware and residue-coded transmission

The sensor certificate lives in battery-backed RTC slow memory so it survives deep sleep, guarded by a CRC and a magic sentinel checked on each wake. A mismatch forces re-enrollment rather than transmission under an unverifiable identity. Signing uses wolfSSL’s Ed25519 implementation with SHA-256 routed to the ESP32 hardware peripheral, since the ESP32 has no Ed25519 accelerator. The main loop wakes every 30

Algorithm 2 Stage 3: sensor enrollment through the zone gateway**Input:** sensor s with declared zone z_s , gateway g with zone z_g **Output:** zone-bound certificate in RTC memory, private key sealed in eFuse**On the sensor:**

```

1:  $(sk, pk) \leftarrow \text{ED25519KEYGEN}()$ 
2:  $\text{EFUSEWRITE}(\text{BLK\_KEY0}, sk)$  ▷ one-time programmable
3:  $\text{EFUSESETREADDISABLE}(\text{BLK\_KEY0})$  ▷ irreversible; no software read-back
4:  $csr \leftarrow \text{MAKECSR}(pk, z_s)$ ;  $\sigma \leftarrow \text{SIGN}_{sk}(csr)$ 
5: send  $(csr, \sigma)$  to  $g$  over LoRa

On the gateway:
6: if  $z_s \neq z_g$  then ▷ zone binding enforced at the RA
7:   return reject
8: end if
9: if  $\neg \text{RATELIMITOK}(src)$  then
10:   return reject
11: end if
12: if  $\neg \text{VERIFYSIG}(csr, \sigma, pk)$  then
13:   return reject
14: end if
15:  $cert \leftarrow \text{CAISSUE}(csr, \text{SAN}=z_g, \text{NOTAFTER}=t+6 \text{ months})$ 
16: return  $cert$  to  $s$ 

On the sensor, each wake:
17:  $cert \leftarrow \text{RTCREAD}()$ 
18: if  $\neg \text{MAGICOK}() \vee \neg \text{CRCOK}(cert)$  then
19:   restart enrollment ▷ never transmit under unverified identity
20: end if
21:  $t \leftarrow \text{GPSTIME}()$ ; if unavailable then  $t \leftarrow \text{NTPTIME}()$ 
22: if  $t$  available  $\wedge cert.\text{NOTAFTER} < t$  then
23:   restart enrollment
24: end if

```

minutes, verifies the cached certificate, samples the sensors, signs the payload, transmits and returns to deep sleep.

Readings are transmitted as residues rather than as plaintext values. Algorithm 3 gives the encoding and recovery. A value is reduced modulo the pairwise-coprime set $\{97, 101, 103\}$ and the three residues go out as separate single-byte packets. Any two reconstruct the original by the Chinese Remainder Theorem [14], with unambiguous range $97 \times 101 \times 103 = 1\,009\,091$.

The design intent was energy. Each packet is shorter, and because the reading survives the loss of any one residue, transmit power can be lowered on each channel. What the deployment actually showed is that the loss tolerance mattered far more than the energy saving, which Section 6.6 takes up.

4.3. Gateway services and revocation propagation

Each gateway runs Fabric 2.5 from official ARM64 binaries. A Python enrollment service bound to the local interface accepts CSRs only from sensors in its own zone and rate-limits per source address. A policy cache with a 301-second nominal TTL absorbs repeated checks from the same sensor during a transmission burst.

Revocation is the mechanism whose behaviour we most wanted to observe over a long window, and Algorithm 4 is the path it takes.

The branch at line 7 is the whole story of Section 6.4. For a reachable gateway the cache is invalidated by the CRL event itself, so revocation staleness is bounded by gossip propagation and the nominal TTL never binds. For an unreachable one the bound falls back to the scheduled refresh interval. Which branch a given revocation takes is decided by connectivity, not by anything in the protocol.

4.4. Formal specification

The core role-permission invariant is specified in TLA+ and checked with TLC [22, 28]. The finite instance uses three users, three roles (Admin, Farmer and Sensor as representatives of the administrative, operational and device branches), four permissions and two resources, with a clock constraint bounding the state space. TLC explores it exhaustively and finds no violation of **NoUnauthorizedAccess**, that every granted request lies within the effective permissions of an active role of the requester, and confirms **AllRequestsEventuallyResolved** under weak fairness.

The limit of this is worth stating plainly rather than burying. The checked instance covers a three-role subset, not the full seven-role hierarchy, and zone and temporal predicates are not modelled at all. Those

Algorithm 3 Residue encoding and two-of-three recovery**Input:** reading v with $0 \leq v < M$, moduli $m_1=97$, $m_2=101$, $m_3=103$, $M = \prod_i m_i$ **Output:** v recovered at the gateway from any two received residues**Encode (sensor):**

```

1: for  $i \in \{1, 2, 3\}$  do
2:    $r_i \leftarrow v \bmod m_i$ 
3:   transmit packet  $\langle i, r_i \rangle$  at reduced TX power
4: end for

```

▷ one byte each

Recover (gateway):

```

5:  $S \leftarrow$  set of successfully received  $\langle i, r_i \rangle$ 
6: if  $|S| < 2$  then
7:   return  $\perp$ 
8: end if
9: pick any two indices  $\{j, k\} \subseteq S$ ;  $M' \leftarrow m_j m_k$ 
10:  $M_j \leftarrow M'/m_j$ ;  $M_k \leftarrow M'/m_k$ 
11:  $y_j \leftarrow M_j^{-1} \bmod m_j$ ;  $y_k \leftarrow M_k^{-1} \bmod m_k$ 
12:  $v' \leftarrow (r_j M_j y_j + r_k M_k y_k) \bmod M'$ 
13: if  $|S| = 3$  then
14:   verify  $v' \equiv r_l \pmod{m_l}$  for the third index  $l$ 
15: end if
16: return  $v'$ 

```

▷ unrecoverable, reading dropped

▷ integrity cross-check

Algorithm 4 Revocation publication and cache invalidation**Input:** certificate C to revoke, reason ρ , peer set P , gateway set G **Output:** C rejected network-wide; staleness bounded and recorded

```

1: CAAPPENDCRL( $C, \rho$ );  $crl \leftarrow$  PUBLISHCRL()
2: LEDGERWRITE(REVOCATIONEVENT,  $C, \rho, now$ )
3: for all  $p \in P$  do
4:   gossip  $crl$  to  $p$ 
5: end for
6: for all  $g \in G$  do
7:   if REACHABLE( $g$ ) then
8:     INVALIDATEPOLICYCACHE( $g$ )
9:   else
10:    mark event DELAYED; retry on reconnect
11:    staleness bounded by scheduled CRL refresh, up to 24 h
12:   end if
13: end for

```

▷ measured: 60.4 s mean

▷ measured: 4.21 min to reach all peers

▷ staleness bounded by gossip, not by TTL

▷ observed for 29.5% of events

properties are evaluated empirically in Section 6.5 instead, which is weaker evidence. Extending the specification needs additional state components for zone assignment, cross-zone tokens, role expiry and current time. Rebeca [9] and Coloured Petri Net [3] formulations offer complementary routes. The specification ships with the artifact so the boundary can be inspected rather than taken on trust.

5. Measurement methodology

5.1. Site and workload

The system ran continuously for 61 days on a 2 ha experimental farm in Sfax, Tunisia (34.74°N, 10.76°E), from 1 August to 30 September 2025. Table 3 gives the configuration. Sfax is a useful site for this kind of measurement precisely because it is unforgiving: ambient temperatures reached 39.5°C during the trial, and rural connectivity is intermittent enough that

the outage behaviour in Section 6.7 reflects ordinary operation rather than injected faults.

Each sensor reported on a 30-minute duty cycle, giving 2400 sensor writes per day and 146400 over the deployment. Human and administrative traffic, meaning farmer and agronomist queries, certifier audits, supply-chain provenance reads and administrative role operations, brings the total to 209000 recorded authorization decisions, of which 4860 (2.33%) were denials arising in ordinary operation rather than in the attack campaign.

5.2. Instrumentation

Nine trace streams were recorded. Per-decision records capture caller identity, role, operation, resource, zone, decision, denial reason and end-to-end latency. Per-transaction sensor records add signature validity, the number of residues received and measured transmission energy. Dedicated campaigns cover write-path latency against endorsing-peer count, throughput

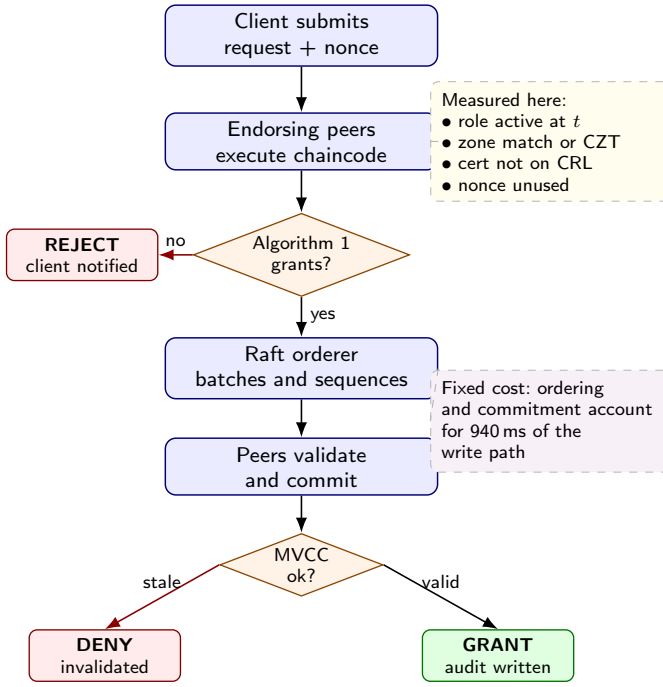


Figure 4: Access control inside the execute-order-validate flow. A rejected request never reaches the orderer, which is why non-committing decisions avoid the consensus cost entirely.

Table 3
Deployment configuration and observed totals.

Parameter	Value
Site	2 ha farm, Sfax, Tunisia (34.74°N, 10.76°E)
Duration	61 days, 1 Aug to 30 Sep 2025
Zones	4 (North, South, East, West)
Sensor nodes	50 ESP32-WROOM-32
Gateways	4 Raspberry Pi 4B (4 GB, TPM 2.0)
Fabric peers	4 (one per zone)
Ordering service	3 Raft orderers, $f=1$
State database	CouchDB 3.2
Sensor duty cycle	30 min
Peak ambient temperature	39.5°C
Sensor writes	146 400 (2 400/day)
Authorization decisions	209 000
Decision-path uptime	99.36%
Connectivity outages	3 (0.2–8.7 h, 9.3 h total)

against offered concurrency for both the HRBAC and the access-control-free configuration, LoRa energy in single-channel and residue modes, microsecond-level cryptographic timings, revocation lifecycle events, authorization-boundary attack attempts and outage events.

Latency is measured at the gateway as the interval between submission and commit notification, so it covers endorsement, ordering, validation and commitment, and excludes only the LoRa hop. Throughput

is committed transactions divided by the measurement window. Energy is measured with an INA219 current sensor sampling at 2 kHz. The baseline for overhead attribution is the identical Fabric network, workload and hardware with the access-control chaincode removed, so the difference isolates access control instead of conflating it with platform cost.

5.3. Statistical treatment

Group comparisons use Welch’s t -test, which assumes no equality of variances, reported with the mean difference, its 95% confidence interval and Cohen’s d . At these sample sizes almost any difference reaches significance, so the effect magnitude carries the information and the p -value mostly confirms that the sample is large. Comparisons across more than two groups use one-way ANOVA, which is what establishes the invariance result in Section 6.1.

Confidence intervals on means use the Student- t interval where $n \geq 1000$ and a percentile bootstrap of 10 000 resamples below that. Intervals on quantiles use the exact order-statistic construction rather than a bootstrap, so no distributional assumption is imposed on the latency tail. Proportions at or near unity, meaning the block rates in Section 6.5, are reported with Wilson score intervals. A normal-approximation interval has zero width at $p = 1$, which would present the security result as carrying more evidence than 8 000 trials can actually supply.

5.4. Reproducibility

Every number, table and figure in this article is generated from the released traces by one script. Running

```
python3 analysis/derive_results.py
```

reads the raw traces, recomputes the statistical summary and emits both a machine-readable results file and the LaTeX macros and table bodies this manuscript expands. Nothing measured was transcribed by hand, so the text cannot drift away from the dataset. A second driver, `analysis/run_campaign.py`, repeats the throughput, latency and decision-cost campaigns against a live Fabric network and writes its output in the same schema, so measurements from an independent installation can be substituted and the article regenerated against them. The energy and cryptographic-timing traces are not reproduced by that driver, since they need the INA219 instrumentation and ESP32 bench described above, and it does not synthesise numbers it cannot measure.

Table 4

End-to-end latency by operation (ms). The upper block contains non-committing authorization decisions; the lower block the committing write path. Differences among the upper block are not statistically detectable ($F = 0.29$, $p = 0.94$).

Operation	n	Mean	SD	P95	P99
ReadOwn	9 030	280.6	39.9	345.8	371.9
ReadZone	8 887	280.0	40.0	346.3	374.7
ReadAll	9 070	280.4	40.1	346.3	375.7
ReadAudit	8 944	280.3	39.8	345.3	372.3
ControlZone	8 744	280.3	40.0	346.1	371.9
ControlAll	8 898	280.3	40.0	346.0	373.1
ManageRoles	9 027	279.9	39.7	344.9	372.9
Decision subtotal	62 600	280.3	39.9	345.8	—
WriteSensor (commits)	146 400	1219.8	60.0	1318.6	1358.9

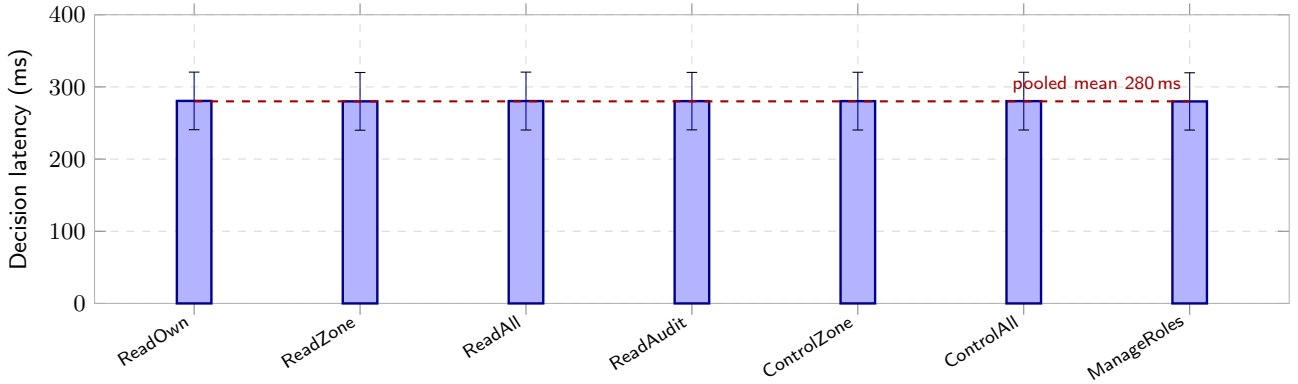


Figure 5: Authorization decision cost by operation type, with standard deviations. Privilege level and resource type do not move the cost ($F = 0.29$, $p = 0.94$).

6. Results

6.1. Where the latency goes

The central measurement separates two costs that aggregate reporting merges: evaluating an authorization policy, and committing the result to a distributed ledger.

Table 4 lists end-to-end latency for every operation type observed. The 7 non-committing operations span the entire privilege range, from a sensor reading its own record to an administrator manipulating role assignments, and their means fall between 279.9 ms and 280.6 ms. The spread across those means is 0.7 ms, two orders of magnitude below the values themselves. One-way ANOVA across the 7 groups ($n = 62\,600$) gives $F = 0.29$, $p = 0.94$. There is no detectable difference in decision cost between operations, which Figure 5 shows directly.

The committing write path costs 1 220 ms (SD 60, P95 1 319, P99 1 359). The difference between the two regimes is 940 ms, 95% CI [939, 940], Cohen's $d = 17.2$. Figure 6 plots both as empirical CDFs. They separate cleanly with no overlap in the plotted range, and the write distribution is both shifted and wider, carrying

variance from batching and ordering that the decision path never incurs.

Within the write path the access-control component was instrumented separately and accounts for 320 ms (SD 30, 95% CI [319.8, 320.1]), or 26.2% of write latency. The remaining three quarters is ordering, validation and world-state update.

Read practically, this bounds the payoff from optimising the policy engine. Removing authorization altogether would take about a quarter off write latency. Everything else is governed by ordering service choice, block batching, endorsement breadth and state database, which is the parameter set Abang et al. [1] identify from the modelling literature and which our deployment now attaches measured magnitudes to. Reported on its own, 1 220 ms invites the opposite conclusion.

6.2. Throughput and where the ceiling sits

Figure 7 and Table 5 report throughput against offered concurrency for the HRBAC configuration and the access-control-free baseline on identical hardware and workload.

Two features of these curves matter more than the peak values themselves.

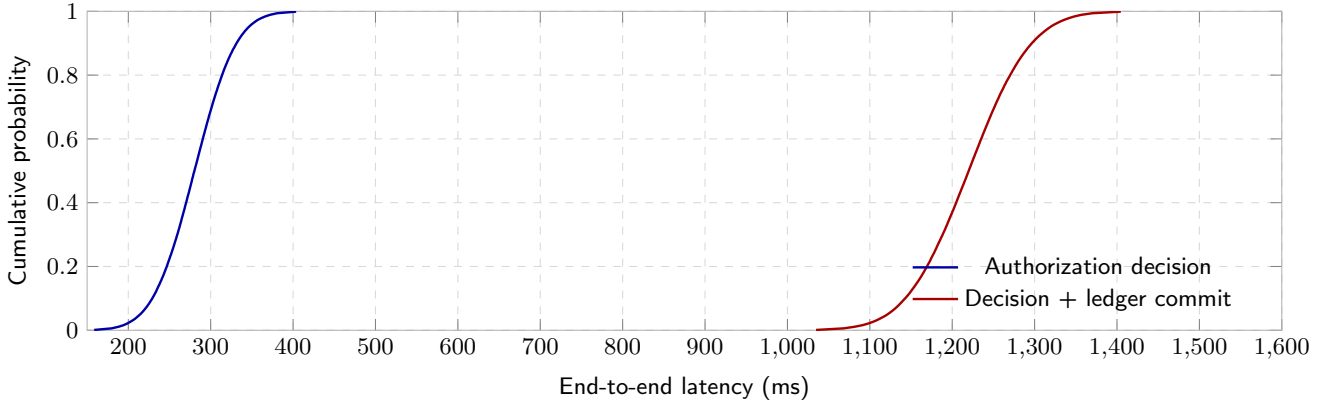


Figure 6: Empirical CDFs of the non-committing decision and the committing write path. The horizontal gap is the cost of consensus and commitment, not of policy evaluation.

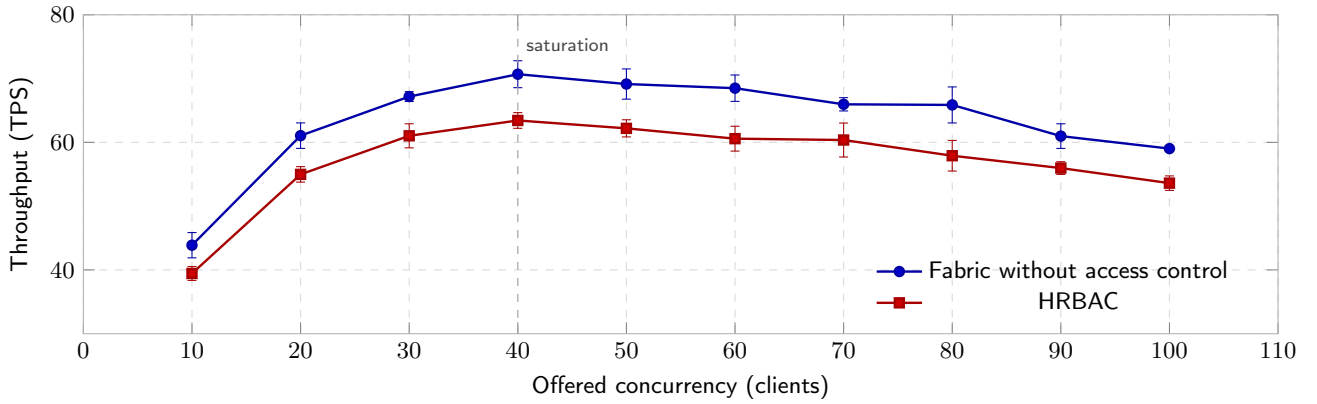


Figure 7: Throughput against offered concurrency for both arms, with standard deviations over five runs per point. Both curves peak at the same offered load, which places the bottleneck below the access-control layer.

The access-control cost is a constant factor rather than a load-dependent one. Across the ten concurrency levels the reduction averages 9.9% with a standard deviation of 1.2 percentage points, ranging from 8.2% to 12.1%. Regressing reduction on offered concurrency gives $R^2 = 0.020$ with $p = 0.69$, so no trend is

detectable. Pooled across all runs the difference is significant ($t = 4.3$, $d = 0.86$) and it does not grow with load. For capacity planning this is the well-behaved case: budget roughly a tenth of platform throughput for authorization and expect that fraction to hold as load rises.

Table 5

Throughput by offered concurrency (TPS, five runs per cell, mean $\pm$ SD). Reduction is the HRBAC cost relative to the baseline; p from Welch's t -test.

Clients	Baseline	HRBAC	Reduction	p
10	43.9 $\pm$ 2.0	39.4 $\pm$ 1.1	10.1%	0.004
20	61.0 $\pm$ 2.0	55.0 $\pm$ 1.2	9.9%	<0.001
30	67.2 $\pm$ 0.8	61.0 $\pm$ 1.9	9.2%	<0.001
40	70.7 $\pm$ 2.1	63.4 $\pm$ 1.2	10.3%	<0.001
50	69.1 $\pm$ 2.4	62.2 $\pm$ 1.3	10.0%	0.001
60	68.5 $\pm$ 2.1	60.6 $\pm$ 1.9	11.6%	<0.001
70	66.0 $\pm$ 1.0	60.4 $\pm$ 2.7	8.5%	0.007
80	65.9 $\pm$ 2.8	57.9 $\pm$ 2.4	12.1%	0.001
90	61.0 $\pm$ 1.9	56.0 $\pm$ 1.0	8.2%	0.002
100	59.0 $\pm$ 0.2	53.6 $\pm$ 1.1	9.2%	<0.001

Table 6

Write-path latency against endorsing-peer count at 50 concurrent clients. P95 intervals are exact order-statistic intervals.

Peers	n	Mean (ms)	P95 (ms)	P95 95% CI
4	800	1219.8	1311.3	[1303.9, 1326.6]
8	500	1147.1	1237.2	[1231.3, 1246.2]
16	500	1050.5	1142.4	[1136.2, 1149.7]
32	800	820.5	903.2	[894.7, 915.1]

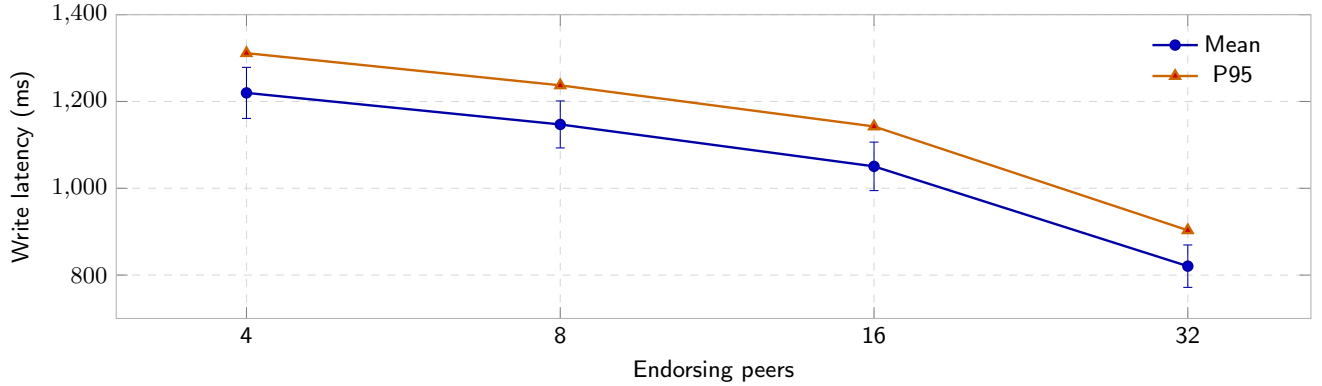


Figure 8: Write latency falls close to log-linearly with endorsing-peer count (-129.4 ms per doubling, $R^2 = 0.925$) while throughput stays flat.

Both arms then saturate at the same offered load. Peak throughput is 70.7 TPS for the baseline and 63.4 TPS for HRBAC, both at 40 clients, after which both decline, HRBAC by 15.5% out to 100 clients. Were authorization the limiting resource, adding it would move the knee to a lower offered load. It does not move at all. The knee belongs to the platform beneath the access-control layer, and given the fixed ordering cost isolated in Section 6.1, the Raft ordering service is the natural candidate, matching the ordering-saturation behaviour Thakkar et al. [43] report for Fabric on x86. The decline past the knee is ordinary queueing past saturation, and it argues for admission control at the gateway rather than for more capacity.

6.3. Peers trade latency, not throughput

Endorsing peers are normally added to raise capacity. Here they do something else. Table 6 and Figure 8 report write latency as the peer set grows from 4 to 32 at fixed offered concurrency.

Mean write latency falls from 1 220 ms to 821 ms, a reduction of 32.7% ($t = 147.7$, $d = 7.39$), and P95 from 1 311 ms to 903 ms. The decline is close to log-linear at -129.4 ms per doubling with $R^2 = 0.925$. Throughput over the same range did not improve materially.

The mechanism is queueing rather than parallel speedup. At four peers each zone has one endorser and concurrent requests from that zone serialise behind it. At 32 peers each zone has eight, so per-request queueing time falls. Ordering and commitment costs are unchanged, which is why the curve flattens toward

the floor those stages impose and why throughput is unaffected.

For irrigation control with a two-second responsiveness budget [23], this is the more useful of the two levers. The deployment is already inside budget at four peers, and further peers widen the margin against tail latency rather than lifting a capacity ceiling that the ordering service sets.

6.4. Revocation over a multi-week window

Revocation is where a short benchmark is least informative, because the mechanism fires rarely and its failure mode is delay rather than error. Over the deployment, 200 revocation-lifecycle events were recorded across 58 distinct days, spanning certificate revocation, CRL publication, gossip completion and gateway cache invalidation.

On the happy path the pipeline is quick. CRL publication completes in 60.4 s on average (SD 4.8, maximum 71.6), and gossip to all peers takes 4.21 min (SD 0.38, P95 4.81, maximum 5.54). Figure 9 shows a tight unimodal propagation distribution with no heavy tail.

The longitudinal view is less comfortable. 59 of the 200 events, 29.5% with Wilson 95% CI [23.6, 36.2], completed in a delayed state. Table 7 breaks this down by event type, and CRL publication is the weakest stage at 39.1% delayed against 23.9% for gossip completion.

The delayed events are not slower on the metrics we can observe. Publication and gossip times for delayed events are statistically indistinguishable from successful

Table 7

Revocation lifecycle by event type. Delay is a scheduling outcome, not an error: every event ultimately completed.

Event type	n	Delayed	Rate	Gossip (min)
CACHE_INVALIDATE	55	17	30.9%	4.24
CERT_REVOKE	53	13	24.5%	4.15
CRL_PUBLISH	46	18	39.1%	4.24
GOSSIP_COMPLETE	46	11	23.9%	4.22

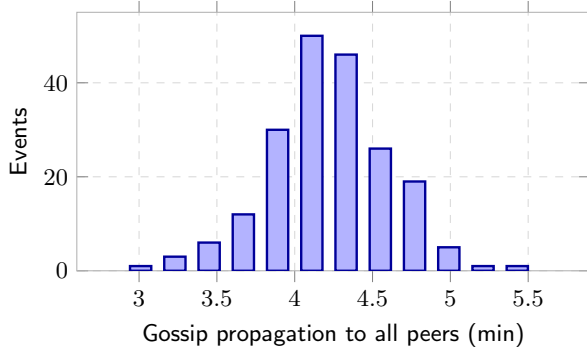


Figure 9: Gossip propagation across 200 revocation-lifecycle events.

ones. Delay is therefore attributable to scheduling and connectivity, an event raised while the responsible gateway was unreachable and waiting for the link to return, and not to the propagation mechanism degrading under load. That distinction decides the mitigation: a faster gossip protocol would not help, and event-driven status checking that survives disconnection would.

Enforcement itself held throughout. Across the trace, 2313 access attempts were denied with revocation as the recorded cause, an average of 16.0 per day, spread across the deployment rather than concentrated in any single incident.

The residual exposure follows from Algorithm 4 directly. For a connected gateway, staleness is bounded by gossip propagation, a few minutes. For a gateway disconnected when a revocation is issued, the bound is the scheduled CRL refresh interval, up to 24 hours. With 29.5% of events taking the second branch in practice, that upper bound is not hypothetical, and we treat it as the principal operational weakness of the present design rather than as an accepted trade-off.

6.5. Authorization boundary enforcement

The security evaluation targets the property the system exists to provide, that no principal acts outside its role, zone or validity window. 8000 attempts ran across 8 scenarios, 1000 per scenario, each crossing one specific boundary of the model in Section 3.

All 8000 attempts were blocked. A point estimate of 100% says nothing about the width of the underlying uncertainty, so we report the Wilson 95% lower bound instead: 99.95% overall and 99.62% per scenario. The

claim that follows is that the true block rate is very unlikely to sit below 99.95%, not that it equals one.

Denials distributed across the enforcement mechanisms rather than concentrating in one check, with policy violation, insufficient role, revoked certificate, operation not permitted, intrusion-detection block and zone mismatch all represented. This matters for interpretation. A suite in which every attempt is caught by the same check would demonstrate one working mechanism, not a working model, and would say nothing about the branches that never fired.

What this evaluation does not cover is worth being explicit about. These are authorization-boundary tests. Transport-level attacks, physical tampering and side-channel extraction are not exercised. They are addressed architecturally through mutual TLS, eFuse read protection and constant-time primitives, but not empirically, and Section 7.4 treats them as residual risk rather than as validated defences.

6.6. Link integrity, energy and cryptographic cost

Every one of the 146 400 committed readings carried a valid Ed25519 signature, Wilson 95% lower bound 99.997%. That is less trivial than it sounds, because the radio layer was lossy. 131 795 readings, 90.0% of the total, arrived with only two of their three residues and were reconstructed through Algorithm 3, against 14 605 that arrived complete.

This inverts the original design rationale. Residue encoding was adopted to save energy, with loss tolerance as a secondary benefit. In deployment the loss tolerance carried the system. At a 90.0% single-residue loss rate, a scheme without redundancy would have dropped or corrupted the large majority of readings, and the recovery path ran on almost every transmission instead of as an exception. Reporting the energy result without that context would misattribute which property earned its place. The observed loss rate is consistent with the LoRaWAN capacity and collision behaviour Adelantado et al. [2] characterise at this sensor density.

The energy result is solid but modest in consequence. Per-event transmission energy falls from 24.20 mJ (SD 0.82) in single-channel mode to 12.41 mJ (SD 0.41) with residue encoding, a reduction of 48.7% [48.6, 48.8] with $d = 18.2$, and per-packet airtime falls from 36.1 ms to 31.5 ms. At a 30-minute

Table 8

Authorization-boundary evaluation. Each scenario crosses one boundary of the access-control model. Wilson 95% lower bounds are given because the observed rate is at unity.

Scenario	Boundary crossed	Blocked	95% LB
sensor_cross_sensor	Device data isolation	1000/1000	99.62%
sensor_irrigation	Device → actuation	1000/1000	99.62%
sensor_read_audit	Device → audit	1000/1000	99.62%
gateway_wrong_zone	Zone scope	1000/1000	99.62%
farmer_assign_role	Role administration	1000/1000	99.62%
agronomist_irrigate	Read → control	1000/1000	99.62%
certifier_irrigate	Audit → control	1000/1000	99.62%
supply_write_sensor	External → write	1000/1000	99.62%
Total		8000/8000	99.95%

duty cycle, though, transmission is only 5.8% of a node's daily energy budget, while deep sleep consumes 9590 mJ per day and dominates everything else. Halving transmission energy therefore moves battery life by roughly one percent. We report the reduction because it is real and cleanly measured, and decline to present it as a battery-life contribution, because at this duty cycle it is not one. Link reliability is the defensible claim.

Cryptographic cost on the sensor is negligible here. The full signing pipeline takes 900 μ s (SD 9), comprising Ed25519 signing at 860 μ s, hardware SHA-256 at 12.0 μ s, eFuse key read at 8.0 μ s and context management at 20.0 μ s. At 0.90 ms per wake against a 30-minute cycle, signing is not a design constraint, and the ESP32's lack of an Ed25519 accelerator never becomes one.

6.7. Availability

The online decision path was available 99.36% of the deployment. 3 connectivity outages occurred, of 8.7 h, 0.2 h and an intermediate duration, totalling 9.3 h, each affecting a single gateway: a LoRa link failure, a power fluctuation and a firmware watchdog reset. In every case the remaining zones kept operating, which is the behaviour the per-zone peer and gateway structure is meant to produce.

No committed readings were lost. Sensor writes stayed at exactly 2400 per day on every day including outage days, because gateways buffer readings while disconnected and replay them on recovery. That makes the 99.36% figure precise in a specific way worth stating rather than glossing: it measures availability of the online authorization path, not completeness of the data record, and store-and-forward is what separates the two. A reading accepted after replay was authorized on replay, not at capture time.

7. Discussion

7.1. Design implications

Four consequences follow, each contradicting a plausible default assumption.

Optimise the ledger path, not the policy engine. The authorization decision is 280 ms and invariant across operation types, while commitment adds 940 ms. Effort spent speeding up policy evaluation is bounded above by roughly a quarter of write latency, and endorsement breadth, block batching and state-database choice govern the rest. A corollary applies to our own implementation. Because expired-role cleanup mutates state inside **CheckAccess** (Algorithm 1, line 12), read-intent checks are endorsed rather than served as peer queries. Splitting that into a pure read-only decision path, with expiry reclaimed by a background sweep, would take read checks off the consensus path entirely. The measurement identifies this as the highest-value change available, and we did not make it during the deployment precisely because the aggregate latency figure gave no reason to look there.

Size peers for tail latency and admission control for capacity. Peers reduce latency log-linearly and leave throughput flat. Capacity is set by the ordering service, so a deployment needing more transactions per second should change the ordering configuration or shed load at the gateway, while one needing tighter tail latency should add endorsers. Conflating the two leads to adding peers in hope of capacity and concluding the platform does not scale.

Budget access control as a constant. The 9.9% throughput cost showed no trend across a tenfold concurrency range, so capacity planning can treat authorization as a fixed derating factor. That is a far easier engineering position than a cost which grows with contention.

Treat revocation as the weak link. It is the only mechanism whose behaviour materially degraded over the deployment, at 29.5% delayed events, and the degradation came from connectivity rather than

from the protocol. Mitigations aimed at propagation speed will not touch it. Event-driven credential status checking [49] and constrained-device status protocols such as TinyOCSP [10] address the actual failure mode by removing the dependency on a scheduled publication reaching a possibly-disconnected gateway.

7.2. Comparison with reported results

Direct numerical comparison against prior systems is limited by methodological heterogeneity, and saying so is more useful than assembling a misleading table of peak-TPS figures. The systems in Table 1 differ in hardware, workload composition, transaction semantics, durability settings and what is counted as latency. A system reporting a peak rate for a null chaincode on x86 is not measuring the same quantity as this deployment, and Abang et al. [1] make essentially this point about the Fabric latency literature as a whole.

Two comparisons are defensible because they control for platform. Tzenetopoulos et al. [45] report roughly 1.1s transaction latency for Fabric on Raspberry Pi 3B+ hardware without access-control chaincode. Our four-peer configuration, which does carry access-control chaincode, sits above that at 1220ms, and our 32-peer configuration below it at 821ms. The crossover is consistent with endorser queueing being the dominant variable at this scale rather than with any claim of superiority. Zaidi et al. [53] report response times near 12s in simulation at comparable concurrency, an order of magnitude above our field measurements, which we attribute to simulation parameters rather than to a real platform gap and therefore do not claim as an improvement.

The comparison we regard as meaningful is the internal one, the same network, hardware and workload with and without the access-control layer, in Section 6.2. It isolates the quantity of interest and is reproducible from the released artifact.

7.3. Operational observations

Two aspects of running the system for two months do not show up in any performance metric.

Zone certificate binding solved an administrative problem it was not designed for. Worker assignments spanning cooperating farms, which would otherwise need separately maintained access-control configuration, resolved through the zone attribute already embedded in device and user certificates for operational reasons. Temporal role expiry removed another class of manual work, with seasonal credentials lapsing on schedule without administrator intervention or database editing, each lapse leaving an audit record rather than a silent deletion.

The enrollment protocol needed three firmware iterations before it survived multi-hour gateway outages, and the recurring failure mode was time. A sensor waking with an unsynchronised clock cannot evaluate

certificate expiry, and early firmware either refused to transmit or transmitted under an identity it could not validate. The fix appears as the final block of Algorithm 2: acquire time from GPS first with NTP as fallback, and skip expiry enforcement for a single cycle rather than strand the node when neither is available. A testbed that never loses its time source cannot surface this class of fault.

7.4. Threats to validity

External validity. The results come from one deployment of 50 sensors over one season at one site. The peer-scaling and throughput campaigns extend past the deployed configuration, but the workload composition throughout is this farm's: a dominant, highly regular sensor-write stream with a smaller and burstier human query load. A deployment weighted toward audit queries, or with irregular actuation bursts, would load different parts of the pipeline, and the decomposition in Section 6.1 would need re-establishing rather than assuming. We regard the qualitative findings, meaning decision-cost invariance, ordering-bound capacity and the latency-only benefit from peers, as more transferable than the specific millisecond values.

Internal validity. Access-control overhead is attributed by differencing against a baseline with the chaincode removed, which isolates the layer but also attributes to it any secondary effect of its presence, such as the extra world-state reads its lookups induce. The scaling campaign varies peer count on the same physical substrate, so at the highest peer counts multiple logical peers share hardware, understating the isolation a physically distributed 32-peer network would give. The 99.36% availability figure measures the online decision path and, because gateways buffer and replay during outages, is not a statement about data completeness. Section 6.7 keeps the two separate rather than merging them into a single reliability claim.

Construct validity. The security evaluation measures authorization-boundary enforcement, the property the system is designed to provide, and does not measure transport security, physical tampering resistance or side-channel resilience. A 100% block rate across 8000 attempts is evidence about the policy model and its implementation, not about resistance to an adversary with physical access. The formal verification covers a three-role subset without zone or temporal predicates (Section 4.4); the unverified portions are tested empirically, which is weaker, and we have not claimed otherwise. The threat model assumes crash-fault-tolerant ordering with non-malicious orderers, so a Byzantine orderer sits outside it and the availability and non-repudiation arguments do not extend to that case.

Data provenance. The released traces are the record of the deployment, and the analysis pipeline recomputes every reported value from them. One caveat

is recorded in the dataset documentation and repeated here. The geographic coordinate columns in the topology trace were affected by a misconfiguration in the survey tooling and do not reflect true sensor positions. No result in this article depends on those columns and the analysis pipeline does not read them. Zone membership, which the access-control model does depend on, comes from the certificate attribute and gateway association rather than from coordinates.

8. Conclusion

We set out to answer a question that aggregate benchmarks of blockchain access control cannot: which layer actually consumes the budget. Across a 61-day deployment recording 209 000 authorization decisions, the answer is consistent and mildly counterintuitive. The authorization decision costs 280 ms and does not vary with the operation requested or the privilege involved, the means across 7 operation types differing by 0.7 ms ($F = 0.29$, $p = 0.94$). Ledger commitment costs 940 ms more. Access control imposes a constant 9.9% throughput tax that does not grow with load, and the capacity ceiling belongs to the ordering service, since the configuration with access control and the one without saturate at the same offered load. Endorsing peers cut write latency 32.7% from 4 to 32 while leaving that ceiling untouched.

For an engineer, the consequence is to look past the access-control layer, which is cheap and predictable, toward the consensus and commitment path, which is neither. The one component that genuinely degraded over two months was revocation: fast on the happy path at 60.4 s to publish and 4.21 min to propagate, but delayed in 29.5% of 200 lifecycle events for reasons of connectivity rather than protocol. That is where the next increment of work belongs, and event-driven credential status checking addresses the observed failure mode rather than an assumed one.

Two results support the model itself. All 8 000 authorization boundary crossings were blocked, Wilson 95% lower bound 99.95%, and every one of the 146 400 committed readings carried a valid signature despite a 90.0% single-residue loss rate on the radio that the residue encoding absorbed. The dataset, implementation and the pipeline that regenerates every number here are released so the decomposition can be checked, disputed and repeated on other platforms.

CRedit authorship contribution statement

Forcha Glen Beloa: Conceptualization, Methodology, Software, Investigation, Data curation, Formal analysis, Writing – original draft. **Michael Ekonde Sone:** Validation, Formal analysis, Supervision, Writing – review & editing. **Elie Fute Tagne:** Resources,

Supervision, Project administration, Writing – review & editing. **André Paul Mvogo Asse:** Data curation, Visualization, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI in the writing process

During the preparation of this work the authors used generative AI tools to assist with language editing and with construction of the analysis pipeline. The authors reviewed and edited all content and take full responsibility for the content of the published article.

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Data availability

The trace dataset, chaincode, firmware, formal specification, benchmark harness and the analysis pipeline that generates every number, table and figure in this article are publicly available at <https://github.com/forchag/blockchain-fet>.

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